

Lazy Loading

Lazy loading is a technique in React where components are loaded **only when they are needed**, instead of loading everything at the beginning.

Normally, when a React app starts, all components are loaded together. This makes the app slower, especially if the app is large.

Lazy loading solves this problem by splitting the code into smaller parts and loading them **on demand**.

Simple definition:

Lazy loading means **loading components only when the user needs them**

1. Benefits of Lazy Loading

Lazy loading gives many advantages in real-world applications.

1. Faster Initial Load

Only important components are loaded first.
So the app opens faster.

2. Better Performance

Less code is loaded at once, so the browser works faster.

3. Reduced Bundle Size

The main JavaScript file becomes smaller.

4. Better User Experience

Users don't have to wait for the entire app to load.

5. Saves Network Data

Only required files are downloaded.

2. When to Use Lazy Loading?

Lazy loading should be used in the right places.

Use Lazy Loading when:

- Large applications
- Multiple pages (routing apps)

- Heavy components (charts, videos, maps)
- Rarely used pages
- Admin dashboards

Avoid Lazy Loading when:

- Small components
- Very frequently used components
- Simple UI elements (buttons, headers)

3. Performance Difference

Without Lazy Loading:

- Entire app loads at once
- Large bundle size
- Slow initial loading
- Poor performance on slow internet

With Lazy Loading:

- Only required components load
- Smaller bundle size
- Faster startup time
- Better performance

Example Comparison:

Feature	Without Lazy Loading	With Lazy Loading
Load Time	Slow	Fast
Bundle Size	Large	Small
Performance	Lower	Better
User Experience	Poor	Good

4. Real-Time Examples

1. E-commerce Website

- Home page loads first
- Cart page loads only when clicked

- Payment page loads later

Improves speed and user experience

2. Video Streaming App

- Home page loads fast
- Video player loads only when user clicks video

Saves memory and improves performance

3. Admin Dashboard

- Charts and analytics load only when needed
- Heavy libraries are not loaded initially

4. Login System

- Login page loads first
- Dashboard loads after login

5. Disadvantages of Lazy Loading

Even though lazy loading is useful, it has some drawbacks.

1. Initial Delay for Component

When user opens a lazy component, there may be a small delay.

2. More Complexity

Code becomes slightly more complex to manage.

3. Need for Loading UI

You must use loading indicators (like “Loading...”).

4. SEO Issues (sometimes)

Lazy loading may affect SEO if not handled properly.

5. Extra Requests

More network requests are created for loading chunks.

6.Example Code

```
import React, { Suspense } from "react";
```

```
const About = React.lazy(() => import("./About"));
```

```
function App() {  
  return (  
    <Suspense fallback={<h2>Loading...</h2>}>  
      <About />  
    </Suspense>  
  );  
}
```

```
export default App;
```